

Math 550
Homework 1

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Exercise 1. Show that $\mathcal{B}_{\mathbf{R}}$ is generated by each of the following:

- the open intervals: $\mathcal{E}_1 = \{(a, b) \mid a < b\}$,
- the closed intervals: $\mathcal{E}_2 = \{[a, b] \mid a < b\}$,
- the half-open intervals: $\mathcal{E}_3 = \{(a, b] \mid a < b\}$ or $\mathcal{E}_4 = \{[a, b) \mid a < b\}$,
- the open rays: $\mathcal{E}_5 = \{(a, \infty) \mid a \in \mathbf{R}\}$ or $\mathcal{E}_6 = \{(-\infty, a) \mid a \in \mathbf{R}\}$,
- the closed rays: $\mathcal{E}_7 = \{(a, \infty) \mid a \in \mathbf{R}\}$ or $\mathcal{E}_8 = \{(-\infty, a) \mid a \in \mathbf{R}\}$.

Proof.

□

Exercise 2. Let $\mathcal{M}$ be an infinite σ -algebra.

- $\mathcal{M}$ contains an infinite sequence of disjoint sets.
- $\text{card}(\mathcal{M}) \geq \aleph_1$.

Proof. a. Using the Axiom of Choice, pick a family of subsets $\{E_n\}_{n \in \mathbf{N}}$ from $\mathcal{M}$. Then:

$$\left(E_n \setminus \bigcup_{k=1}^{n-1} E_k \right)_n$$

is an infinite sequence of disjoint sets. Indeed, let $F_n := E_n \setminus \bigcup_{i=1}^{n-1} E_i$ and consider $F_i \cap F_j$ for natural numbers $i < j$. If $x \in F_i$, then $x \in E_i$. But this means $x \in \bigcup_{k=1}^{j-1} E_k$, hence $x \notin F_j$.

b. Let $F_n := E_n \setminus \bigcup_{k=1}^{n-1} E_k$. Define $f : \mathcal{P}(\mathbf{N}) \rightarrow \mathcal{M}$ by $f(A) = \bigcup_{i \in A} F_i$. Suppose $f(A) = f(B)$. Then $\bigcup_{i \in A} F_i = \bigcup_{i \in B} F_i$. Let $x \in A$. Then $F_x \subset \bigcup_{i \in A} F_i$, hence $F_x \subset \bigcup_{i \in B} F_i$. But since each F_i is disjoint, it must be that $x \in B$. An identical argument shows that if $y \in B$, then $y \in A$, hence $A = B$. Since f is injective, we have $\text{card}(\mathcal{P}(\mathbf{N})) \leq \text{card}(\mathcal{M})$; i.e., $\aleph_1 \leq \text{card}(\mathcal{M})$. □

Exercise 3. An algebra $\mathcal{A}$ is a σ -algebra if and only if $\mathcal{A}$ is closed under countable increasing unions.

Proof. If $\mathcal{A}$ is a σ -algebra, then $\mathcal{A}$ is closed under countable unions, hence it is closed under countable *increasing* unions. Conversely, suppose $\mathcal{A}$ is closed under countable increasing unions. Let $\{E_n\}_{n=1}^{\infty}$ be a family of sets in $\mathcal{A}$. Define:

$$\begin{aligned} F_1 &= E_1, \\ F_2 &= E_1 \cup E_2, \\ &\vdots \\ F_n &= E_1 \cup E_2 \cup \dots \cup E_n. \end{aligned}$$

Then clearly $F_1 \subset F_2 \subset F_3 \dots$. Moreover, it is clear that $\bigcup_{n=1}^{\infty} E_n \subset \bigcup_{n=1}^{\infty} F_n$. Conversely, let $x \in \bigcup_{n=1}^{\infty} F_n$. Then $x \in F_i$ for some $i \in \mathbf{N}$. By construction, $x \in E_1 \cup \dots \cup E_i$, hence $x \in E_j$ for some $j < i$. Whence $x \in \bigcup_{n=1}^{\infty} E_n$, giving the inclusion $\bigcup_{n=1}^{\infty} F_n \subset \bigcup_{n=1}^{\infty} E_n$. Together, we have $\bigcup_{n=1}^{\infty} E_n = \bigcup_{n=1}^{\infty} F_n \in \mathcal{A}$. Since $\{E_n\}_{n=1}^{\infty}$ was an arbitrary family of sets in $\mathcal{A}$, we can conclude $\mathcal{A}$ is a σ -algebra. $\square$